
NAME

RDKitRemoveDuplicateMolecules.py - Remove duplicate molecules

SYNOPSIS

```
RDKitRemoveDuplicateMolecules.py [--infileParams <Name,Value,...>] [--mode <remove or count>] [
--outfileParams <Name,Value,...> ] [--overwrite] [--useChirality <yes or no>] [-w <dir>] [-o <outfile>]
-i <infile>
```

```
RDKitRemoveDuplicateMolecules.py -h | --help | -e | --examples
```

DESCRIPTION

Identify and remove duplicate molecules based on canonical SMILES strings or simply count the number of duplicate molecules.

The supported input file formats are: SD (.sdf, .sd), SMILES (.smi., csv, .tsv, .txt)

The supported output file formats are: SD (.sdf, .sd), SMILES (.smi)

OPTIONS

-e, --examples

Print examples.

-h, --help

Print this help message.

-i, --infile <infile>

Input file name.

--infileParams <Name,Value,...> [default: auto]

A comma delimited list of parameter name and value pairs for reading molecules from files. The supported parameter names for different file formats, along with their default values, are shown below:

SD: removeHydrogens,yes,sanitize,yes,strictParsing,yes

SMILES: smilesColumn,1,smilesNameColumn,2,smilesDelimiter,space,
smilesTitleLine,auto,sanitize,yes

Possible values for smilesDelimiter: space, comma or tab.

-m, --mode <remove or count> [default: remove]

Specify whether to remove duplicate molecules and write out filtered molecules to output files or or simply count the number of duplicate molecules.

-o, --outfile <outfile>

Output file name.

--outfileParams <Name,Value,...> [default: auto]

A comma delimited list of parameter name and value pairs for writing molecules to files. The supported parameter names for different file formats, along with their default values, are shown below:

SD: compute2DCoords,auto,kekulize,yes,forceV3000,no

SMILES: smilesKekulize,no,smilesDelimiter,space, smilesIsomeric,yes,
smilesTitleLine,yes,smilesMolName,yes,smilesMolProps,no

Default value for compute2DCoords: yes for SMILES input file; no for all other file types.

--overwrite

Overwrite existing files.

-u, --useChirality <yes or no> [default: yes]

Use stereochemistry information for generation of canonical SMILES strings to identify duplicate molecules.

-w, --workingdir <dir>

Location of working directory which defaults to the current directory.

EXAMPLES

To remove duplicate molecules and generate output files containing unique and duplicate SMILES strings, type:

```
% RDKitRemoveDuplicateMolecules.py -i Sample.smi -o SampleOut.smi
```

To remove duplicate molecules without using stereochemistry information for generation of canonical SMILES and generate output files containing unique and duplicate SMILES strings, type:

```
% RDKitRemoveDuplicateMolecules.py -u no -i Sample.sdf -o SampleOut.sdf
```

To count number of unique and duplicate molecules without generating any output files, type:

```
% RDKitRemoveDuplicateMolecules.py -m count -i Sample.sdf
```

To remove duplicate molecules from a CSV SMILES file, SMILES strings in column 1, name in column 2, and generate output SD files containing unique and duplicate molecules, type:

```
% RDKitRemoveDuplicateMolecules.py --infileParams  
"smilesDelimiter,comma,smilesTitleLine,yes,smilesColumn,1,  
smilesNameColumn,2" --outfileParams "compute2DCoords,yes"  
-i SampleSMILES.csv -o SampleOut.sdf
```

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SEE ALSO

RDKitConvertFileFormat.py, RDKitRemoveInvalidMolecules.py, RDKitRemoveSalts,
RDKitSearchFunctionalGroups.py, RDKitSearchSMARTS.py, RDKitStandardizeMolecules.py

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The functionality available in this script is implemented using RDKit, an open source toolkit for cheminformatics developed by Greg Landrum.

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